

Part-A

Q1). Decision stump Prediction.

Rule: Predict "+" if sneezing = Yes, else "-".

Dataset:- 4 Examples

correct predictions:- 3

Incorrect predictions:- 1

Training error rate formula

$$\text{error rate} = \frac{\text{Number of Incorrect Predictions}}{\text{Total Predictions}} = \frac{1}{4}$$

$$= 0.25 \text{ (or) } 25\%$$

Memorized model:- Predicts each training example perfectly $\rightarrow$ error = 0%.

Q2). Training error as splitting criterion for each feature, calculate the error after splitting:

split on Age (x_1):

Young $\rightarrow$ 2 examples $\rightarrow$ both correct

mid $\rightarrow$ 2 examples $\rightarrow$ both correct

old $\rightarrow$ 2 examples $\rightarrow$ 1 incorrect.

Error rate:

$$\text{error} = \frac{1}{6} = 16.7\%$$

Split on exercise (X_2):

High $\rightarrow$ 2 examples $\rightarrow$ both correct

Medium $\rightarrow$ 2 examples $\rightarrow$ 1 incorrect

Low $\rightarrow$ 2 examples $\rightarrow$ both correct

Error rate:

$$\text{error} = \frac{1}{6} = 16.7\%$$

Split on Diet (X_3):

Poor $\rightarrow$ 4 examples $\rightarrow$ 1 incorrect

Good $\rightarrow$ 2 examples $\rightarrow$ both correct

Error rate

$$\text{error} = \frac{1}{6} = 16.7\%$$

All splits yield the same error rate

$\rightarrow$ No clear winner

Q3) Entropy & information gain

Initial entropy

$$H(S) = P+ \log_2 P + - P- \log_2 P- = -0.5 \log_2 0.5 \\ = -0.5 \log_2 0.5 = 1.$$

Split on exercise (X_2):

High $\rightarrow$ 2 examples $\rightarrow$ both yes $\rightarrow$ entropy = 0

Medium $\rightarrow$ 2 examples $\rightarrow$ 1 yes, 1 NO $\rightarrow$ entropy = 1

Low $\rightarrow$ 2 examples $\rightarrow$ both NO $\rightarrow$ entropy = 0

weighted entropy

$$H_{\text{split}} = \frac{2}{6} (0) + \frac{2}{6} (1) + \frac{2}{6} (1) = \frac{1}{3}$$

Information Gain

$$IG = H(S) - H_{\text{split}} = 1 - \frac{1}{3} = \frac{2}{3} \approx 0.66$$

Q4) Confusion matrix metrics.

Metric	Formula	Value
Accuracy	$TP + TN / \text{Total}$	$25 + 55 / 100 = 0.8$
Precision	$TP / TP + FP$	$25 / 25 + 15 = 0.625$
Recall	$TP / TP + FN$	$25 / 25 + 5 = 0.833$
Specificity	$TN / TN + FP$	$55 / 55 + 15 = 0.786$
F1 score	$\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	0.714

Q5). Distance calculation (KNN)

Points A(5,4)

use euclidean distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$A(2,4) :- \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9} = 3$$

$$B(4,4) :- \sqrt{(5-4)^2} = \sqrt{1} = 1$$

$$C(4,6) :- \sqrt{(5-4)^2 + (4-6)^2} = \sqrt{1+4} = \sqrt{5} = 2.24$$

1-NN Prediction: closer is B $\rightarrow$ Blue

3-NN Prediction: Nearest neighbors are A (red), B (Blue), C (red) $\rightarrow$ Majority = red.

Q6). k-fold cross-validation.

Mean error formula

$$\text{mean error} = \frac{\text{sum of fold errors}}{\text{No. of folds.}}$$

$$k=1 : \frac{0.2+0.25+0.15+0.3}{4} = 0.225$$

$$k=3 : \frac{0.15+0.2+0.1+0.2}{4} = 0.1625$$

$$k=4 : \frac{0.1+0.15+0.1+0.2}{4} = 0.1375$$

Best Generalization $k=5$ (lowest error).